ClinoKG: Constructing a Medical Knowledge Graph from Biomedical Literature

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Abstract

ClinoKG constructs a medical Knowledge Graph from PubMed Abstracts. It adapts RelForge to biomedical language, extracting relations among drugs, diseases, genes, and symptoms from two million abstracts. The resulting facts are stored in a Relational Fact Lattice layout for fast lookup and provenance-aware inspection. ClinoKG produces about 1.2 million entities and 6 million relations, with approximately 91% precision in a manually checked sample. The work focuses on Knowledge Graph Construction rather than question answering, providing a structured foundation for downstream clinical search and verification.

Keywords

Medical knowledge graph, PubMed Abstracts, Relation extraction, Biomedical facts

1. Introduction

Biomedical literature contains a large, changing inventory of entities and relations. Knowledge Graph Construction can make that information searchable, but generic extraction models often misread abbreviations and nested biomedical mentions. ClinoKG adapts RelForge as described in [1] and adopts the Relational Fact Lattice storage idea from [2]. The evaluation is intentionally limited to graph quality and does not use a question-answering dataset.

2. Method

ClinoKG applies a biomedical version of RelForge to 2 million PubMed Abstracts. Candidate entities are detected with span boundaries and normalized to canonical forms using type-specific dictionaries. For each pair, the extractor predicts a typed relation and records the sentence, abstract identifier, confidence, and character offsets. A post-processing stage merges synonyms only when type and context agree. The Knowledge Graph is then written in a Relational Fact Lattice layout: facts are bucketed by subject and relation, with secondary indexes for object and source lookup. This storage format supports rapid neighborhood queries while preserving provenance. A manually reviewed sample is stratified by relation type and entity category. We compare relation precision with a dictionary-based baseline that fires when curated terms co-occur in a sentence.

3. Experiments

ClinoKG produces 1.21 million entities and 6.04 million relations from 2 million PubMed Abstracts. Manual review estimates 91.0% relation precision, compared with 76.8% for the dictionary-based baseline. Lookup of a subject-relation bucket takes 3.4 ms at the 95th percentile. The most common errors involve speculative language and relations spanning multiple sentences.

System	Relation precision	Entities / relations
Dictionary baseline	76.8%	--
ClinoKG	91.0%	1.21M / 6.04M

4. Conclusion

ClinoKG demonstrates that RelForge can support high-precision biomedical Knowledge Graph Construction at million-abstract scale. The Relational Fact Lattice layout enables fast lookup with provenance preserved. Better handling of speculation and cross-sentence evidence remains necessary before clinical use.

References

[1] Raman, P. (2020). RelForge: Neural Relation Extraction for Automatic Knowledge Graph Construction. Conference on Natural Language Systems (CNLS).

[2] Raman, P. (2022). Relational Fact Lattices for Grounding Compact Clinical Text Generators. Clinical Informatics Letters (CIL).